

Knowledge and Language 2025/2026

literature Review

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I. ABSTRACT

This literature review explores the integration of Knowledge Graphs (KGs) and Retrieval-Augmented Generation (RAG) techniques for domain-specific question answering, the challenge we are trying to tackle is building a system that can reliably and accurately answer complex, domain-specific questions about Portuguese cuisine and other recipes using the food.com dataset [1]. We will use draws RecipeRAG [1], a KG-driven recipe generation framework, as reference for the methodologies to build a food chatbot. Definitions and concepts are supported by the course powerpoint on semantic web framework (RDF, OWL, SPARQL), establishing the link between symbolic and knowledge graph representation.

II. CONNECTIONS WITH THE TOPICS COVERED BY THE COURSE

Large Language Models (LLMs) demonstrate strong generalization but fail in domain-specific reasoning due to three main limitations: (1) lack of contextual understanding, (2) incomplete factual knowledge, and (3) weak recall of precise facts [?]. This motivates the integration of symbolic knowledge (RDF/OWL-based graphs) with neural inference (LLMs and RAG) to ensure both reasoning and factual grounding. RDF can be serialized in formats such as N-TRIPLES, XML, JSON-LD, or Turtle, ensuring interoperability across systems. Vocabularies define terms and relationships and ontologies define classes, hierarchies, and property restrictions. OWL (Web Ontology Language) can be used with RDF to express richer logical constraints, such as domain/range restrictions and class relations.

This symbolic foundation aligns with the three “knowledge-aware” dimensions of KG-augmented NLP methods outlined in [?]:

- KG-Augmented Retrieval (3.1.1): The use of structured queries (SPARQL) or textualized triples to retrieve relevant information for LLMs, comparable to RAG pipelines [Lewis et al., 2020].
- KG-Augmented Reasoning and Controlled Generation (3.1.2–3.1.3): Incorporating graph reasoning steps (symbolic inference or logical validation) within the generation loop.
- Knowledge-Aware Training (3.2): Using KG-derived data to fine-tune LLMs, injecting factual consistency.

- Knowledge-Aware Validation (3.3): Employing SPARQL or rule-based checks to verify the factual correctness of generated answers.

This emphasizes the distinction between symbolic reasoning (FOL-based) and neural reasoning, emphasizing that symbolic models provide interpretability and factual precision, while neural models capture correlations and language variability.

III. KNOWLEDGE GRAPH CONSTRUCTION/ENRICHMENT

In RDF, knowledge is represented through triples — for example:

`<Bacalhau_a_Brás> <hasIngredient> <Egg>`

Each triple can be enriched through RDFS with class and property hierarchies. Ontology design (defined in OWL) allows specifying property restrictions, such as `onlyUsesIngredient` or `hasMaxPrepTime`. In [2] RecipeKG is constructed as a large-scale recipe knowledge graph combining ingredients, cooking techniques, and regions, and enriching it with semantic annotations. Enabling structured querying and semantic interoperability through URIs and ontologies. Symbolic KGs are inherently sparse, so Knowledge Graph Embeddings (KGEs) are employed to map entities and relations into continuous vector spaces. Foundational models [2] include TransE, RotatE, and QuatE, representing relations as translations, rotations, and quaternion transformations respectively [Han et al., 2023].

IV. KNOWLEDGE RETRIEVAL VIA NLP

Knowledge retrieval connects natural language input to structured or embedded KG representations. SPARQL is the declarative query language for RDF graphs [?], enabling semantic retrieval using triples and filters. However, natural language interfaces require NLP-based query interpretation, extracting entities and intents (via tools like spaCy or BERT) and mapping them to KG entities (URIs). KGE-based retrieval [2] significantly outperforms text-based retrieval when handling multiple user constraints. As noted: “Performance declined as the number of criteria increased... Nevertheless, the results underscore the effectiveness of KGEs in supporting multi-criteria recipe retrieval, while simultaneously revealing the limitations of purely text-based embeddings.”77

V. SPARQL QUERY GENERATION

SPARQL queries operate over RDF graphs, and their construction can be made in 3 different ways:

- The user using the Chatbot need to input the SPARQL query directly or use a form-based interface that generates the query based on user selections.
- LLMs can be fine-tuned or prompted to generate SPARQL queries from natural language questions, using few-shot examples or templates to guide the generation.
- NLP tools like semantic parsers can convert natural language into SPARQL queries by identifying entities, intent, relations, restrictions and query structure.

VI. RETRIEVAL-AUGMENTED GENERATION (RAG)

RAG, first formalized by [?] as a hybrid model that retrieves external documents as "the input sequence x to retrieve text documents z and use them as additional context when generating the target sequence y ". This approach separates retrieval from generation, improving factuality and reducing hallucinations. While the original RAG used text documents, later adaptations like the mentioned RecipeRAG [2] replace text retrieval with KG-based retrieval.

RecipeRAG describe their RAG pipeline as follows:

"We chose RotatE, the strongest-performing KGE model, to bootstrap the recipe generation process. The system retrieves the top-ranking recipes based on aggregated relevance scores, selects the top 5, and feeds them—together with user criteria—into the generator (DeepSeek-R1-0528)."

This pipeline combines the retrieval mechanism of [?] with structured KG enrichment, turning symbolic knowledge into context for generative models. Given this, we will implement the following layered knowledge representation model: symbolic layer (RDF/OWL) $\rightarrow$ Query Layer (SPARQL) $\rightarrow$ neural layer (LLM generation given top 5 most similar recipes).

RAG thus operationalizes the Knowledge-Aware Inference paradigm [?], where external structured knowledge is used to enhance reasoning and factual correctness.

VII. INTEGRATING CONCEPTS FOR THE PROPOSED ARCHITECTURE

The proposed architecture is:

KG Construction:

Build a Portuguese recipe KG using RDF/OWL.

Define Recipe, Ingredient, Region, and Technique as OWL classes with property restrictions (hasIngredient only Ingredient).

Entity Recognition:

Use spaCy/BERT for entity linking and intent recognition.

Map to KG URIs using ontology matching and synonyms.

SPARQL Verification:

Run queries to validate retrieved facts (e.g., prep time, allergens).

RAG Generation:

Feed top-5 recipes + structured SPARQL facts into an LLM.

Use prompt templates and tests (for example "jeopardy questions generation task) inspired by [?] for performance metrics evaluation.

Validation:

Post-generation KG validation (checking logical consistency via OWL reasoning).

VIII. CONCLUSION AND FUTURE WORK

The reviewed literature converges on the same insight: combining symbolic reasoning (RDF, OWL, SPARQL) and neural reasoning (KGE, RAG, LLMs) yields systems that are both factual and context-aware. Empirically, RecipeRAG [Loesch et al., 2025] and RAG [Lewis et al., 2020] demonstrate that KG-driven retrieval enhances generation quality, while the taxonomy of [Agrawal et al., 2024] clarifies how such systems reduce hallucination through Knowledge-Aware Inference, Learning, and Validation.

Together, these works guide the design of the CuocoGPT chatbot: a Linked Data-based KG, queried via SPARQL, and integrated with a RAG-enhanced LLM, achieving both accuracy and interpretability.

REFERENCES

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- [2] T. Loesch, Y. Zhang, and A. Kumar, "Reciperag: A knowledge graph-driven approach to personalized recipe retrieval and generation," in *Proceedings of RAGE-KG 2025*, 2025.